$\boxed{1}$ Given,

(i) For CPU 1,

Clock rate $= 2.7\,GHz = 2.7 \times 10^9\,Hz$

CPI's are 7, 2, 3 and 6

For CPU 2,

Clock rate $= 3\,GHz = 3 \times 10^9\,Hz$

CPI's are 5, 4, 2, 1

Instruction count $= 10^6$

For CPU 1,

$$\text{Average } CPI_1 = \frac{\{(0.2 \times 7) + (0.5 \times 2) + (0.2 \times 3) + (0.1 \times 6)\} \times 10^6}{10^6}$$

$$= 3.6$$

For CPU 2,

$$\text{Average } CPI_2 = \frac{\{(0.2 \times 5) + (0.5 \times 4) + (0.2 \times 2) + (0.1 \times 1)\} \times 10^6}{10^6}$$

$$= 3.5$$

$\therefore (CPI_1 - CPI_2) = (3.6 - 3.5) = 0.1$

$\therefore$ On average, CPU1 takes 0.1 more clock cycles than compared to CPU2.

(2) For CPU1,

$$\text{Execution Time}, T_1 = \frac{\text{Instruction Count} \times \text{Average CPI}}{\text{Clock Rate}}$$

$$= \frac{10^6 \times 3.6}{2.7 \times 10^9}$$

$$= 1.33 \times 10^{-3} s$$

For CPU 2,

$$\text{Execution Time}, T_2 = \frac{10^6 \times 3.5}{3 \times 10^9} = 1.16 \times 10^{-3} s$$

$$\text{Difference} = (T_2 - T_1) = 1.7 \times 10^{-4} s$$

$\boxed{2}$ Given, For CPU 1,

Clock rate $= 4$ GHz $= 4 \times 10^9$ Hz

CPI $= 4$

For CPU2,

Clock rate $= 2$ GHz $= 2 \times 10^9$ Hz

$$IPS_1 = \frac{4 \times 10^9}{4} = 1 \times 10^9 \text{ IPS} \quad IPS_1 > IPS_2$$

$$IPS_2 = \frac{2 \times 10^9}{6} = 3.33 \times 10^8 \text{ IPS} \quad \therefore \text{CPU 1 performs better.}$$